```python
from hmmlearn import hmm
import numpy as np

model = hmm.CategoricalHMM(n_components=3)

model.startprob_ = np.array([0.5, 0.3, 0.2])
model.transmat_ = np.array([
    [0.6,0.3,0.1],
    [0.3,0.4,0.3],
    [0.2,0.3,0.5]
])
model.emissionprob_ = np.array([
    [0.7,0.3],
    [0.4,0.6],
    [0.2,0.8]
])

# Correct observation format
obs = np.array([[0],[1],[1],[0]])

print("Hidden states:", model.predict(obs))
```

output:

```
Hidden states: [0 0 0 0]
```